Exercise 3.7

1) $\dfrac{8.765 \times \sqrt[3]{27.03}}{24.51}$

$$x = \frac{8.765 \times \sqrt[3]{27.03}}{24.51}$$

$$\lg x = \lg \frac{8.765 \times 27.03^{\frac{1}{3}}}{24.51}$$

$$= \lg 8.765 + \lg 27.03^{\frac{1}{3}} - \lg 24.51$$

$$= \lg 8.765 + \frac{1}{3} \times \lg 27.03 - \lg 24.51$$

$$= 0.9427 + \frac{1}{3} \times 1.4319 - 1.3894$$

$$= 0.9427 + 0.4773 - 1.3894$$

$$= 1.4200 - 1.3894$$

$$= 0.0306$$

$$\lg x = 0.0306$$

$$x = \text{Antilg } 0.0306$$

$$= 1.073$$

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b) $x = \dfrac{\sqrt{9.18} \times 8.02^2}{9.83}$

$\lg x = \lg \dfrac{9.18^{\frac{1}{2}} \times 8.02^2}{9.83}$

$= \lg 9.18^{\frac{1}{2}} + \lg 8.02^2 - \lg 9.83$

$= \frac{1}{2} \times \lg 9.18 + 2 \times \lg 8.02 - \lg 9.83$

$= \frac{1}{2} \times 0.9628 + 2 \times 0.9042 - 0.9926$

$= \cancel{\frac{1}{2}} \times 0.4814 + 1.8084 - 0.9926$

$= 2.2898 - 0.9926$

$\lg x = 1.2972$

$x = \text{Antilg } 1.2972$

$x = 19.82$ //

c) $x = \dfrac{\sqrt{0.0945} \times 4.821^2}{48.15}$

$\lg x = \lg \dfrac{0.0945^{\frac{1}{2}} \times 4.821^2}{48.15}$

$= \lg 0.0945^{\frac{1}{2}} + \lg 4.821^2 - \lg 48.15$

$= \frac{1}{2} \times \lg 0.0945 + 2 \times \lg 4.821 - \lg 48.15$

$= \frac{1}{2} \times \bar{2}.9754 + 2 \times 0.6831 - 1.6825$

$= \bar{1}.4877 + 1.3662 - 1.6825$

$= \bar{1}.1714$

$x = \text{Antilg } \bar{1}.1714$

$= 0.1483$ //

$$-2+1$$
$$2\overline{)1.2544}\quad 0.6272$$
$$\bar{1}.8762 \times 2 = \bar{1}.7524$$

d) $x = \dfrac{3 \times 0.752^2}{\sqrt{17.96}}$

$\lg x = \lg \dfrac{3 \times 0.752^2}{17.96^{\frac{1}{2}}}$

$= \lg 3 + 2 \times \lg 0.752 + \frac{1}{2} \times \lg 17.96$

$\cancel{= 0.4771 + \bar{1}.8762 \times 2}$

$= 0.4771 + 2 \times \bar{1}.8762 + \frac{1}{2} \times 1.2544$

$= 0.4771 + \bar{1}.7524 + 0.6272$

$x = \bar{1}.6023$

$x = \text{Antilg } \bar{1}.6023$

$= 0.4002$ //

e) $x = \dfrac{6.591 \times \sqrt[3]{0.0782}}{0.9821^2}$

$\lg x = \lg \dfrac{6.591 \times 0.0782^{\frac{1}{3}}}{0.9821^2}$

$1 + \bar{1}$
$1 + -1$
$1 - 1 = 0$

$= \lg 6.591 + \frac{1}{3} \times \lg 0.0782 - 2 \times \lg 0.9821$

$3\overline{)\bar{3}+\bar{2}.18932}\quad \bar{1}.631$

$= 0.8190 + \frac{1}{3} \times \bar{2}.8932 - 2 \times \bar{1}.9921$

$= 0.8190 + \bar{1}.6311 - \bar{1}.9842$

$0.8190 + \bar{1}.6311 = 0.4501$

$= 0.4501 - \bar{1}.9842$

$= 0.4659$

$\bar{1}.9921 \times 2 = \bar{1}.9842$

$x = \text{Antilg } 0.4659$

$= 2.923$ //

$0.4501 - \bar{1}.9842 = 0.4659$

$-1 - -1$
$-1 + 1 = 0$

f) $x = \dfrac{3.251 \times \sqrt[3]{0.0234}}{0.8915}$

$\lg x = \lg \dfrac{3.251 \times 0.0234^{\frac{1}{3}}}{0.8915}$

$= \lg 3.251 + \lg 0.0234^{\frac{1}{3}} - \lg 0.8915$

$= \lg 3.251 + \frac{1}{3} \times \lg 0.0234 - \lg 0.8915$

$= 0.5120 + \frac{1}{3} \times \bar{2}.3692 - \bar{1}.9501$

$\bar{1}.4564$

$= 0.5120 + \bar{1}.4564 - \bar{1}.9501$

$x = 0.0183$

$x = \text{Antilg } 0.0183$

$= 1.043$